Project: Worked examples
Title: Accumulator sizing (gas law)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The pre-charge gas volume V0 of a hydraulic accumulator that delivers, or absorbs, a liquid volume ΔV between the lowest working pressure p1 and the highest p2, from $p \cdot V^n$ = const with absolute pressures and an exponent n between 1 (isothermal) and 1.4 (adiabatic); the pre-charge pressure at the charging temperature; the next size from your own size list; and the bar·litre product against the PSSR Schedule 1 250 bar·L exception. Three modes: energy storage, shock and pulsation (screening only) and emergency stroke (ΔV is the stroke volume, linkable from the cylinder circuit). Not a pressure-vessel code check. Absolute pressures: p_a = p_gauge + p_atm (or as entered, in Absolute mode) Gas volume: V0 = ΔV / [(p0/p1)^(1/n) − (p0/p2)^(1/n)], n = 1 isothermal … 1.4 adiabatic, then × the real-gas factor f_rg (user input) Pre-charge rules (records): p0 ≤ r_0·p1 (default 0.9, bladder type) and p2/p0 ≤ r_p (default 4) Pre-charge at the charging temperature: p0(T_c) = p0(T_max)·(T_c + 273.15)/(T_max + 273.15) Next size: the smallest listed size not below V0 Bar·litres: PS·V, PS = p2 as a gauge pressure, V the selected size (else V0); compared with 250 bar·L (PSSR Schedule 1, indication only)

Ref: Pressure Systems Safety Regulations 2000 (SI 2000/128) Schedule 1 — 250 bar·litre exception, indication only
Ref: Pressure Equipment Directive 2014/68/EU and UK PER 2016 — Pressure equipment; not checked here
Ref: Esposito, Fluid Power with Applications Ch. on accumulators — Gas-charged accumulator sizing, $p \cdot V^n$ = const

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Duty	Energy storage		assumption: Slow clamp-and-hold cycle: stored-energy duty.	any
dV	Liquid volume ΔV delivered, absorbed or (emergency stroke) needed for the stroke	14	L	spec: Hydraulic specification HS-12: clamp stroke plus leakage make-up, 14 L per cycle	0.001 … 1000 L
p_basis	Pressures entered as	Absolute		assumption: Pressures entered as absolute bar, as the source numbers are quoted.	any
p_0	Gas pre-charge pressure p0, at the maximum operating temperature	90	bar	spec: Hydraulic specification HS-12: pre-charge 90 bar at the maximum oil temperature	0 … 700 bar
p_1	Lowest working pressure p1	100	bar	spec: Hydraulic specification HS-12: lowest clamp pressure 100 bar	0 … 700 bar

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_2	Highest working pressure p2	210	bar	spec: Hydraulic specification HS-12: pump cut-out 210 bar	0 … 700 bar
p_atm	Atmospheric pressure (gauge to absolute)	1.01325	bar	assumption: Standard atmosphere, derived constant.	0.5 … 1.1 bar
n	Gas exponent n (1 isothermal, slow; 1.4 adiabatic, fast)	1		assumption: Isothermal: the clamp holds for minutes between cycles.	1 … 1.4
f_rg	Real-gas factor on V0 (1 = ideal gas; from a gas-property source above 200 bar)	1		assumption: Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.	0.8 … 1.5
T_max	Maximum operating temperature (p0 applies here)	50	degC	spec: Hydraulic specification HS-12: oil temperature limit 50 °C	-40 … 150 degC
T_c	Charging temperature	20	degC	assumption: Charged in the workshop at 20 °C.	-40 … 100 degC
r_0	Pre-charge rule: p0 at most this fraction of p1 (absolute)	0.9		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	0.1 … 1
r_p	Pre-charge rule: p2/p0 at most this ratio (absolute)	4		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	1 … 20
sizes	Available accumulator sizes (from the manufacturer's range)	4 row(s)		reference: record: illustrative size list invented for this example, to be replaced by the manufacturer's range	any
sizes[1]		V = 10 L			
sizes[2]		V = 20 L			
sizes[3]		V = 32 L			
sizes[4]		V = 50 L			

3. Assumptions

A1. All gas-law pressures are absolute. A gauge entry is converted with the atmospheric pressure entered here, so the same numbers read as gauge or as absolute give different volumes. ($p \cdot V^n$ = const on a fixed mass of gas.)
A2. The exponent n is the user's: 1 for a slow discharge with heat exchange (isothermal), towards 1.4 for a fast one (adiabatic). The larger n gives the larger volume for a given ΔV. (Gas-law exponents, closed form.)
A3. The gas is ideal unless a real-gas factor is entered; above 200 bar the sheet warns that nitrogen departs from the ideal-gas law and a factor from a gas-property source should be entered. (Real-gas correction as a user factor.)
A4. The pre-charge rules (p0 ≤ 0.9·p1 and p2/p0 ≤ 4 by default) are records for the accumulator type and make and are checked as PASS/FAIL; the right values come from the manufacturer. (Rules are records, not built-in limits.)
A5. The bar·litre product uses the highest working pressure as a gauge pressure and the selected size (else the calculated volume). It indicates whether the 250 bar·L exception of PSSR Schedule 1 can apply; it is not a compliance statement, and PED / UK PER, PSSR examination and the vessel's own design code stay with the manufacturer and the competent person. (Addendum B §B5 regulatory gating: no pressure-vessel code check.)
A6. mode (Duty) is taken as Energy storage. (Slow clamp-and-hold cycle: stored-energy duty.)
A7. p_basis (Pressures entered as) is taken as Absolute. (Pressures entered as absolute bar, as the source numbers are quoted.)
A8. p_atm (Atmospheric pressure (gauge to absolute)) is taken as 1.01325 bar. (Standard atmosphere, derived constant.)
A9. n (Gas exponent n (1 isothermal, slow; 1.4 adiabatic, fast)) is taken as 1. (Isothermal: the clamp holds for minutes between cycles.)
A10. f_rg (Real-gas factor on V0 (1 = ideal gas; from a gas-property source above 200 bar)) is taken as 1. (Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.)
A11. T_c (Charging temperature) is taken as 20 degC. (Charged in the workshop at 20 °C.)
A12. Energy storage duty: ΔV is the liquid volume the accumulator delivers between p_2 and p_1 over a cycle; n = 1 for a slow discharge, towards 1.4 for a fast one. (Gas-law sizing for a stored-energy duty.)

4. Working

```
[14] pa_0 = p_0
= 90 bar
= 90 bar
```
Pre-charge pressure, absolute
```
[15] pa_1 = p_1
= 100 bar
= 100 bar
```
Lowest working pressure, absolute
```
[16] pa_2 = p_2
= 210 bar
= 210 bar
```
Highest working pressure, absolute
```
[17] V0_ideal = dV / [(pa_0/pa_1)^(1/n) - (pa_0/pa_2)^(1/n)]                p·V^n = const
= 14 L / [(0.9) - (0.42857)]
= 29.697 L
```
Gas volume for the displacement, ideal gas
```
[18] V0 = V0_ideal * f_rg
= 29.697 L × 1
= 29.697 L
```
Pre-charge gas volume V0 (accumulator size required)

```
[19] T_ratio = (T_c / 1 K) / (T_max / 1 K)
= 20 degC / 1 K / (50 degC / 1 K)
= 0.90716
```
Charging over maximum absolute temperature
```
[20] p0 = pa_0 * T_ratio
= 90 bar × 0.90716
= 81.645 bar
```
Absolute pre-charge pressure at the charging temperature
```
[21] p0_set = p0
= 81.645 bar
= 81.645 bar
```
Pre-charge to set at the charging temperature (absolute)
```
[22] u_r0 = pa_0 / (r_0 * pa_1)
= 90 bar / (0.9 × 100 bar)
= 1
```
Pre-charge against the p0/p1 rule
```
[23] u_rp = (pa_2 / pa_0) / r_p
= 210 bar / 90 bar / 4
= 0.58333
```
Pressure ratio against the p2/p0 rule
```
[24] V_sel = smallest listed size not below V0
= V0 = 29.697 L, 4 sizes listed
= 32 L
```
Selected accumulator size
```
[25] u_V = V0 / V_sel
= 29.697 L / 32 L
= 0.92803
```
Required volume over the selected size
```
[26] V_ref = V_sel
= 32 L
= 32 L
```
Volume used for the bar·litre product (selected size)
```
[27] PS = max(pa_2 - p_atm, 0 bar)
= max(210 bar - 1.0132 bar, 0 bar)
= 208.99 bar
```
Highest working pressure, gauge
```
[28] bar_litres = PS * V_ref
= 208.99 bar × 32 L
= 6687.6 bar·L
```
Pressure-volume product PS·V
```
[29] BL_pssr = 250 bar*L
= 250 bar·L
```
PSSR Schedule 1

PSSR Schedule 1 exception threshold (indication)

5. Results

SYMBOL	VALUE	DESCRIPTION
V0	29.697 L	Accumulator gas volume required
V_sel	32 L	Selected size
u_V	0.92803	Selected size utilisation
p0	81.645 bar	Pre-charge, absolute, at the charging temperature
p0_set	81.645 bar	Pre-charge to set at the charging temperature

SYMBOL	VALUE	DESCRIPTION
u_r0	1	Pre-charge rule utilisation (p0/p1)
u_rp	0.58333	Pressure ratio utilisation (p2/p0)
bar_litres	6687.6 bar·L	Pressure-volume product (bar·litres)
pssr	"6687.6 bar·L is at or above 250 bar·L: the exception does not apply; PSSR and PED / UK PER duties follow."	PSSR Schedule 1 250 bar·L exception (indication only)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pre-charge within the p0/p1 rule	u_r0 = 1	u_r0 ≤ 1	1	accumulator.rules record
PASS	Pressure ratio within the p2/p0 rule	u_rp = 0.58333	u_rp ≤ 1	0.583	accumulator.rules record

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: r_0, r_p (Pre-charge rules); sizes (Accumulator size list). Enter each as one so its source, edition and licence travel with it.
WARNING — realgas: p_2: the highest working pressure is 210 bar absolute, above 200 bar; nitrogen departs from the ideal-gas law there and the volume shown is the ideal-gas value. Enter a real-gas factor f_rg from a gas-property source.

8. Verification

Template fp.accumulator@1.0.0 Content hash bc8ad9545ba31a3e Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0